Florian Simperl

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Scientific software engineer and applied physicist (PhD, TU Wien) specializing in FAIR scientific data and machine-learning-ready datasets. Implemented NeXus standards at Scienta Omicron and with FAIRmat, contributed to the NXxps schema, and developed XPS data integration for NOMAD. Generated 9 million simulated spectra for Transformer training on HPC and led development of NIST SESSA, with 800 downloads in 2025. Seeking to advance scientific computing for materials discovery and design.

CORE COMPETENCIES

- **Scientific Data Modeling:** NeXus data schemas for materials science, FAIR data for spectroscopy, contributor to the open-source *pynxtools-xps* converter, structured datasets of organic and inorganic compounds for NN training.
- **Scientific Data Engineering:** Python pipelines (NumPy, SciPy, Pandas, h5py, scikit-learn, scikit-image) for experimental and simulated data; large-scale spectroscopic dataset generation on HPC (SLURM, Linux).
- **Materials Science & Surface Analysis:** X-ray photoelectron spectroscopy, electron transport simulations in solids, layered materials investigation, optical properties of EUV photoresists, MLIPs for carbon nanomembranes.
- **Software Engineering & AI Tooling:** maintainable C++/Python codebases, GitHub/GitLab for code development (CI/CD pipelines), cross-platform releases, documentation and tutorials; Claude Code for SESSA development.

WORK EXPERIENCE

Project Assistant
Jun 2026 - Sep 2026

TU Wien, *Vienna, AT*

- **Experimental Data Workflow:** Developed a Python workflow (SciPy, NumPy, Pandas, h5py, scikit-image) and GUI interface (ipywidgets) for low-energy electron microscopy data on graphite: automated denoising, standardisation, quantitative image analysis and extraction of band structure properties.

Project Assistant (PhD Programme)
Mar 2023 - Apr 2026

TU Wien, *Vienna, AT*

- **Training Data at Scale:** Generated about 9 million simulated XPS spectra on the ASC HPC cluster, covering organic and inorganic materials (CRC Handbook and the Materials Project), and built the dataset pipeline for model training.
- **ML on Scientific Data:** Trained Transformer models for automated quantitative XPS analysis (Ray Tune, WandB) and validated them against experimental reference spectra; performed molecular dynamics of carbon nanomembranes using MLIPs for XPS (LAMMPS, TurboGAP), analysed molecular structures with OVITO, ASE and VMD.
- **Scientific Software (NIST SESSA):** Led development of the C++ Monte Carlo package: FFT-based speedup of about 100 times, Python compatibility, FAIR-compliant output; supported users through bug fixes and tutorials.
- **Experimental Data:** Built a Python pipeline at the ELI-ALPS laser facility that converts attosecond time-resolved photoemission measurements into phase maps and photoemission delays, adopted for future beamline users.

Visiting Research Scientist
Oct 2025 - Nov 2025

Paul Scherrer Institute (PSI), *Villigen, CH*

- **Reference Data:** Measured 20 XPS reference spectra of single elements/multi-element compounds for ML model.

R&D Engineer Intern
May 2024 - Nov 2024

Scienta Omicron, *Uppsala, SE*

- **Data Standards:** Implemented NeXus-based data standards for commercial scientific instruments with the FAIRmat consortium; built the interface between Scienta Omicron's software and the NOMAD repository and contributed to *pynxtools-xps*, which converts native instrument data into standardised NXxps records.

Hardware R&D Engineer Intern
Jul 2022 - Jan 2023

MaxWell Biosystems AG, *Zurich, CH*

- **Reliability Testing:** Verified technical requirements of a CMOS-based microelectrode-array product through accelerated lifetime and reliability tests; analysed the results statistically in Python (SciPy, Pandas, Seaborn) and reported them for design and process decisions (Jira, Confluence).

EDUCATION

PhD in Applied Physics
Mar 2023 - Jul 2026

TU Wien, *Vienna, AT*

- **PhD Programme:** Fellow of the European Marie Skłodowska-Curie Actions (MSCA) Network 'EUSpecLab'
- **Thesis Title:** "Machine Learning Algorithms and Computational Methods for Quantitative Photoelectron Spectroscopy"

MSc in Physics
Sep 2020 - Feb 2022

ETH Zurich & University of Toronto, *Zurich, CH / Toronto, CA*

- **Specialisation:** Particle Physics, Machine Learning, Ultrafast Laser Physics
- **Thesis Title:** "Analysing the space charge effect with the General Particle Tracer Code in PIRL-DIVE mass spectrometry for single molecule detection efficiency"

BSc in Physics
Sep 2017 - Sep 2020

ETH Zurich, *Zurich, CH*

SKILLS & TOOLS

Programming	Python, C++, MATLAB, SQL, Bash, LaTeX
Scientific Computing	COMSOL, LAMMPS, TurboGAP, ASE, OVITO, VMD, SLURM
Machine Learning	PyTorch, PyTorch Lightning, Ray Tune, Weights & Biases
Software Development	GitHub/GitLab, CMake, Claude Code, Confluence, Jira
Languages	German (native), English (C1), French (B1), Dutch (A2)
Certified Trainings	SQL for Data Science (2022), Modern C++ Software Design (2024), AI for Science Bootcamp with NVIDIA Modulus, PINNs and FourCastNet (2024), Introduction to Deep Learning (2024)

ADDITIONAL EXPERIENCE

Coordinator, ML4SPEC2025
Jun 2023 - May 2025

WE-Heraeus Seminar "Machine Learning for Spectroscopy", *Brussels, BE*

- **Conference Organisation:** Led the organising committee for an 80-participant international conference: timelines, budget, sponsorship and communication across partner institutions.

SELECTED PUBLICATIONS

- **Simperl, F.** and Werner, W. S. M. (2025). Journal of Applied Physics, 138(15) https://doi.org/10.1063/5.0296600
- Werner, W. S. M., **Simperl, F.**, et al. (2025). Journal of Physical Chemistry C, 129(45) https://doi.org/10.1021/acs.jpcc.5c06405
- Werner, W. S. M., **Simperl, F.**, et al. (2026). Natl Std. Ref. Data Series (NIST NSRDS) https://doi.org/10.6028/NIST.NSRDS.100-2026

SELECTED SOFTWARE & DATASETS

- **NXxps Data Schema:** https://github.com/FAIRmat-NFDI/pynxtools-xps
- **NIST SESSA:** https://www.nist.gov/srd/nist-standard-reference-database-100
- **Transformer Network and Datasets:** https://doi.org/10.48436/eybcx-t0a02
- **CarbOx Code:** https://github.com/Fuzz-Muffin/CarbOx

HOBBIES

Mountaineering, Photography, Ski touring, Saxophone, Trail running, Arduino

REFEREES

Available upon request.